

Domain Term of the Week

Episode 2 : Connected Factory

A Connected Factory is a digitally integrated manufacturing environment where machines, sensors, systems, and people are interconnected through IoT, cloud, and data platforms. It enables real-time monitoring, data-driven decision-making, predictive maintenance, and seamless coordination across production, supply chain, and enterprise systems.

Activities involved in the term



Asset connectivity & integration
Connecting Operations using IoT Infra



Data Acquisition
Capturing real-time data (temperatures, vibrations, throughput, etc.).



Data transmission
Sending data securely via gateways and protocols.



Data storage & management
Storing data in data lakes or warehouses



Advanced analytics
Applying AI/ML models for operations optimization

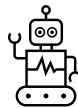
Support Activities



Real-time monitoring: Dashboards and alerts for production status, machine health and KPIs.



System Integration: Regeneration
Integrating factory systems with enterprise applications (e.g., SAP S/4HANA).



Process Automation: Automating workflows and decision-making using integrated systems (MES, ERP)

Benefits to the Organization



Predictive maintenance minimizes unexpected equipment failures



Early defect detection and process standardization.



Data-driven insights enable faster and accurate decisions.



Lower maintenance, energy, and operational costs.



End-to-end tracking from production to delivery.



Increased output through optimized resource utilization